



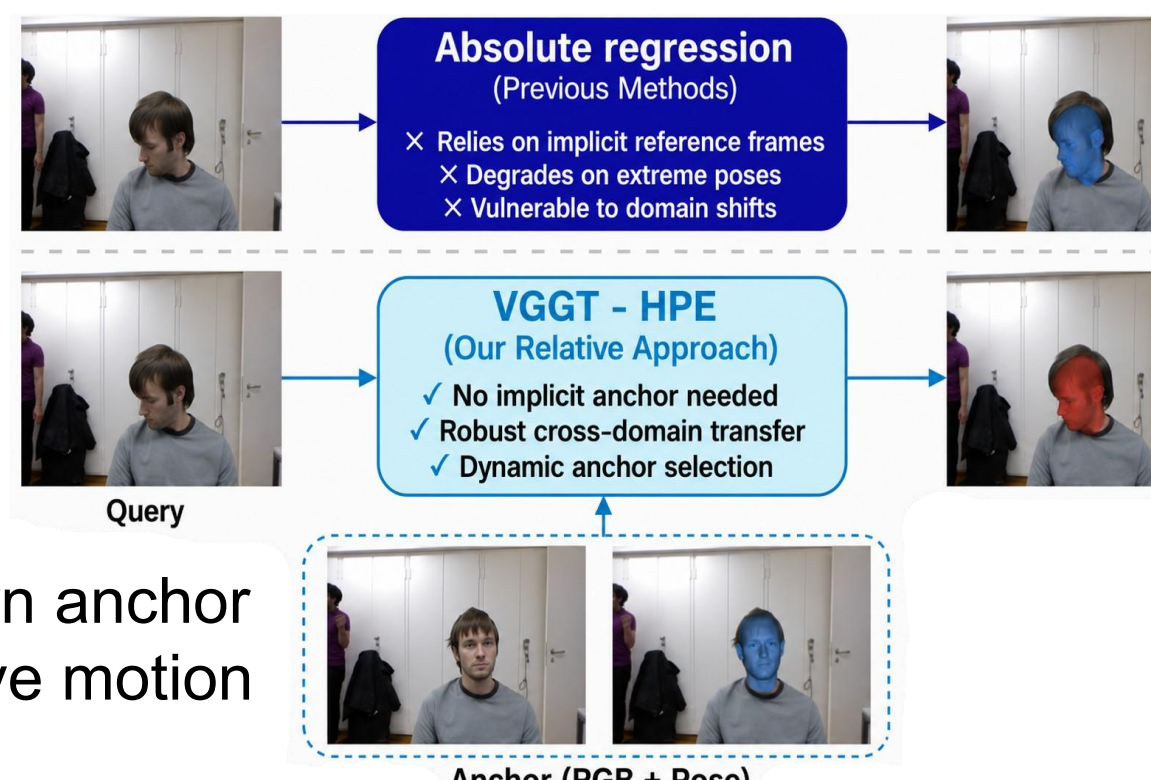
1 Motivation

Old methods: regress pose from one image relative to a hidden canonical frame.

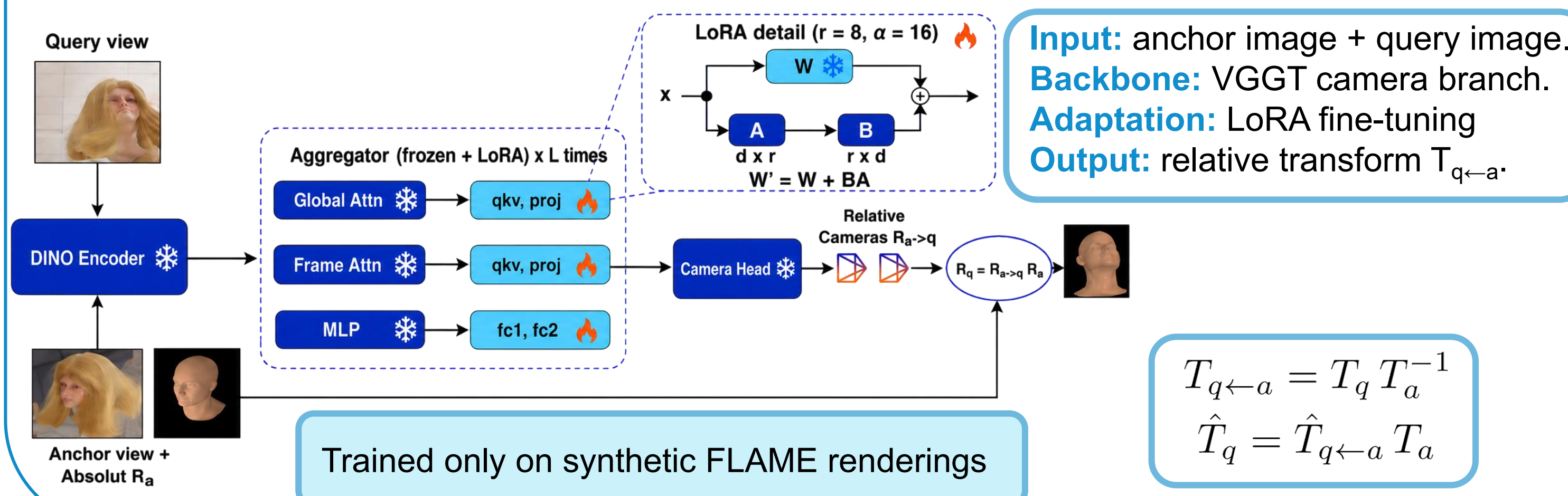
Problem: extreme poses and domain shifts are hard because the reference frame is only learned implicitly.

VGGT-HPE: uses a known anchor image and predicts the relative motion to the query.

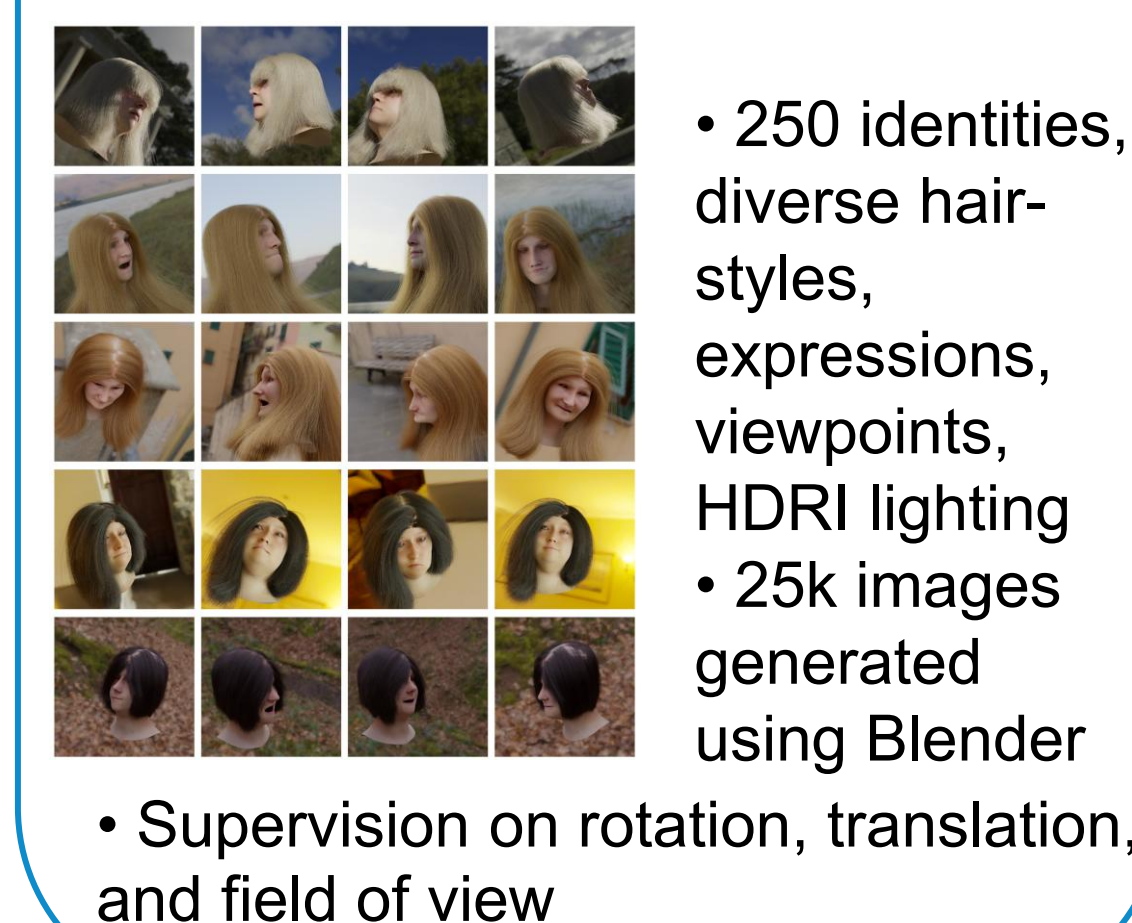
Benefit: choose an easy anchor at test time to control prediction difficulty.



2 Method



3 Synthetic Training Data



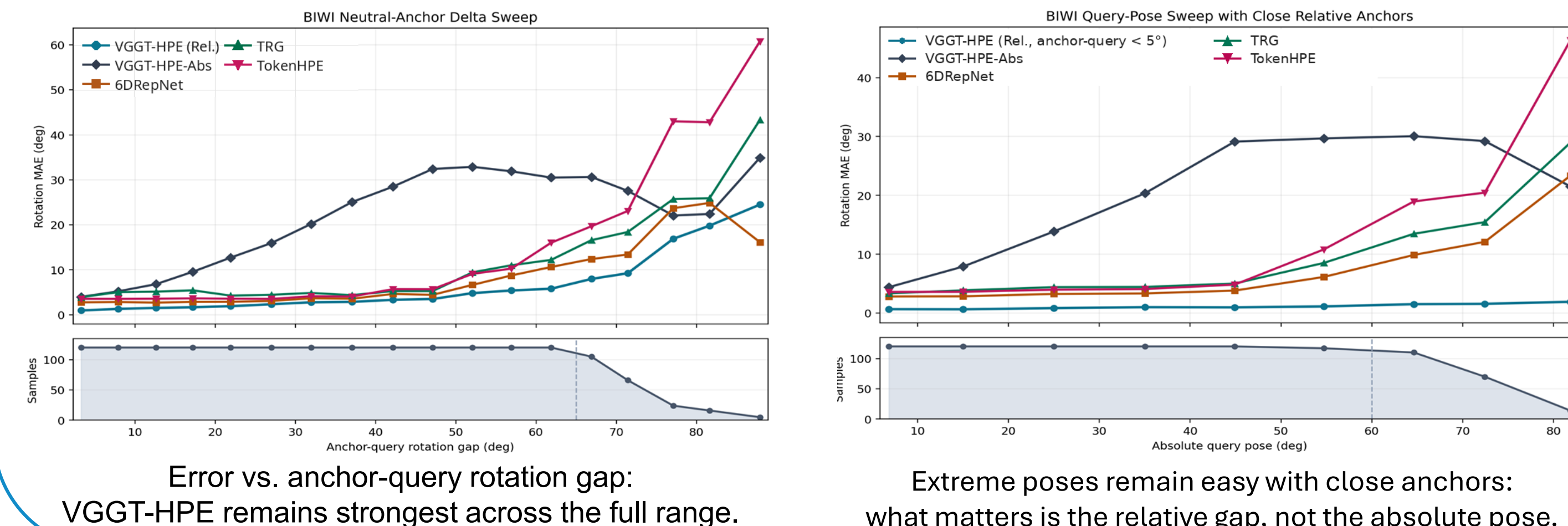
4 Main Results on BIWI

Method	Yaw ↓	Pitch ↓	Roll ↓	MAE ↓	Data
<i>Reported numbers</i>					
6DRepNet	3.24	4.48	2.68	3.47	M
TokenHPE	3.95	4.51	2.71	3.72	M
TRG	3.04	3.44	1.78	2.75	M
VGGT-HPE (Rel., ours)	2.24	3.04	3.17	2.82	S
<i>Reproduced under shared MTCNN protocol</i>					
6DRepNet	3.74	4.95	3.04	3.91	M
TokenHPE-v1	5.57	6.23	3.79	5.20	M
TRG	4.58	7.18	3.68	5.15	M
VGGT-HPE-Abs (ours)	4.90	7.01	3.53	5.15	S
VGGT-HPE (Rel., ours)	2.24	3.04	3.17	2.82	S

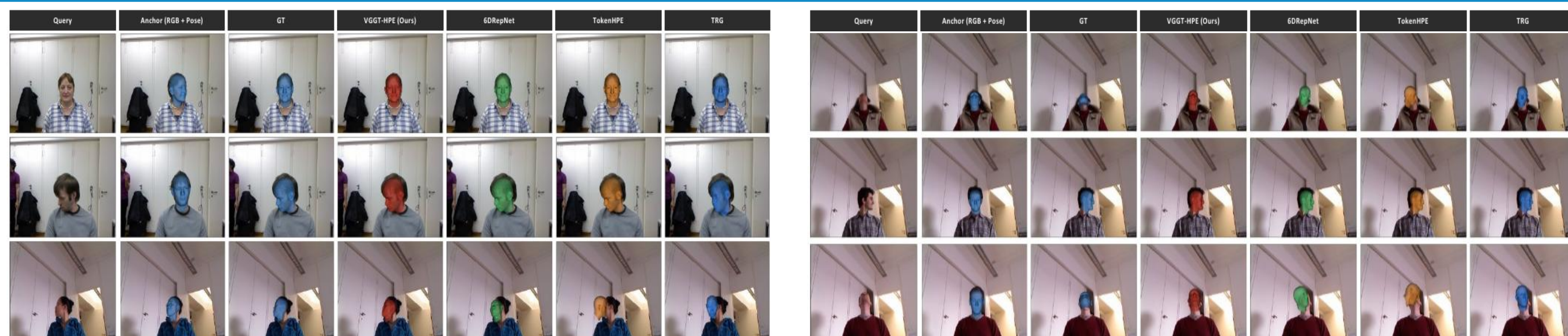
5 Controlled Benchmarks

Hard pairs		Easy pairs	
8.87°		0.96°	
Neutral anchor / Extreme query		Near-neutral anchor / small gap	
Hard benchmark		Easy benchmark	
Method	MAE ↓	Method	MAE ↓
VGGT-HPE-Abs	31.02	VGGT-HPE-Abs	3.98
TokenHPE	22.51	TokenHPE	3.61
TRG	17.23	TRG	3.45
6DRepNet	13.33	6DRepNet	2.80
VGGT-HPE (Rel., ours)	8.87	VGGT-HPE (Rel., ours)	0.96

6 Error Analysis



7 Qualitative Results (BIWI)



8 Ablation

LoRA adapts best: full fine-tuning hurts VGGT priors; head-only overfits synthetic data.

Predicted anchors are effective: using a 6DRepNet anchor achieves **4.08°** on the full BIWI set and **10.26°** on the hard subset.

Full supervision helps: rotation-only gets **3.12°**, while rotation + translation + FoV reaches **2.82°**.

Relative formulation + LoRA + full camera supervision = best cross-domain transfer.

9 Takeaways

- ✓ Relative prediction is easier than absolute regression.
- ✓ The same backbone performs much better in relative mode.
- ✓ VGGT-HPE transfers from synthetic training to real BIWI images.
- ✓ Anchor choice at test time provides controllable difficulty.